

E009

Web Design Fundamentals

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A developer is building a site for a bakery whose brand color is a soft pink. The site gets about 2,000 visits a week. The “Place Order” control is currently a div with a click handler. Which element should the control use?

- A. The same div with a pointer cursor added.
- B. A bold span element.
- C. A button element.
- D. A pink div that matches the brand.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

An outdoor store’s product page shows a photo of a red 30-liter hiking backpack with side water-bottle pockets. The photo was taken by the store’s in-house photographer, and the file is 480 KB. Which alt text is best?

- A. An empty alt attribute.
- B. “backpack_final.jpg”
- C. “Photo by in-house photographer, 480 KB.”
- D. “Red 30-liter hiking backpack with side water-bottle pockets.”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A page uses a purely decorative swirl graphic as a divider between sections. What should its alt attribute be?

- A. Every image needs descriptive alt text.
- B. Leave the alt attribute off entirely.
- C. “Decorative swirl divider image.”
- D. Empty (alt=“”), so screen readers skip it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A designer chose a gray to match the company logo for 16px body text. The page recently won an internal design award. The text-to-background contrast ratio is 3.8:1. Does it meet WCAG Level AA?

- A. No; AA requires 7:1.
- B. No; normal text needs at least 4.5:1.
- C. Yes, because it matches the brand's logo color.
- D. Yes; the minimum is 3:1.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A 24px heading has a contrast ratio of 3.5:1 against its background. Does it meet WCAG Level AA?

- A. Yes; large text needs at least 3:1.
- B. No; all text needs 4.5:1.
- C. Contrast rules never apply to headings.
- D. No; AA requires 7:1.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A 1,200-line stylesheet contains `p { color: black; }`, then `#intro { color: green; }`, and, last in the file, `.note { color: blue; }`. The element is `<p id='intro' class='note'>`. What color is its text?

- A. Green.
- B. Blue, because classes always override IDs.
- C. Blue, because `.note` comes last.
- D. Black, because the element rule applies first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

An element `<p class='note'>` is styled by `p.note { color: red; }` and, later in the file, `.note { color: blue; }`. What color is its text?

- A.** The browser default color.
- B.** Blue.
- C.** Red.
- D.** Blue, because the last rule always wins.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A `.card` rule sets `width: 300px; padding: 20px; border: 2px solid; margin: 16px; background: #ffeccc; and box-sizing: content-box`. How wide is the card from its left border edge to its right border edge?

- A.** 322px
- B.** 344px
- C.** 300px
- D.** 376px

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A box has `width: 300px; padding: 20px; border: 2px solid; and box-sizing: border-box`. How wide is its content area?

- A.** 300px
- B.** 344px
- C.** 278px
- D.** 256px

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A site uses a three-column grid on desktop and is hosted on a fast content delivery network. It has media queries for small screens, but on phones it still looks like a tiny, zoomed-out desktop page. What is most likely missing?

- A.** The viewport meta tag (width=device-width, initial-scale=1).
- B.** More media queries.
- C.** Fixed 1,200px widths on every container.
- D.** A faster hosting provider.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A 40-line script in the page's head calls document.getElementById('menu') to attach a click handler. The menu has six links. The call returns null, though the element exists further down the page. What is the best fix?

- A.** Scripts in the head can never access page elements.
- B.** Add the defer attribute to the script tag (or move it to the end of the body).
- C.** Change the id to a class.
- D.** Use getElementsByClassName instead.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A designer adds outline: none to all buttons and links because the focus ring looks unattractive. User research shows 80% of visitors use a mouse. What should the team do?

- A.** Leave it removed, since most visitors use a mouse.
- B.** Set tabindex='-1' on buttons so they never get focus.
- C.** Replace it with a clear, custom focus style that fits the design.
- D.** Add focus styles to links only.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A developer is building a site for a bakery ~~whose brand color is a soft pink. The site gets about 2,000 visits a week.~~ The “Place Order” control is currently a div with a click handler. Which element should the control use?

Target word: correct element for an action **Decides:** It triggers an action, so it is a button.

A	Dropped element	Looks clickable but still lacks keyboard support and a role.
B	Confusion set	A span is a generic inline element with no control behavior.
C	Correct	Buttons are focusable, respond to Enter and Space, and announce their role.
D	Camouflage	Brand color does not decide the element.

2. Answer D Concept 2

An outdoor store’s product page shows a photo of a red 30-liter hiking backpack with side water-bottle pockets. ~~The photo was taken by the store’s in-house photographer, and the file is 480 KB.~~ Which alt text is best?

Target word: alt text for an informative image **Decides:** The image shows the product the user is deciding to buy.

A	Inverted operation	Marks an informative image as decorative, hiding it from screen readers.
B	Dropped element	A file name carries no useful description.
C	Camouflage	Describes the file, not what the image shows.
D	Correct	Conveys the information the photo gives a sighted shopper.

3. Answer D Concept 2 · transfer

A page uses a purely decorative swirl graphic as a divider between sections. What should its alt attribute be?

Target word: alt text for a decorative image **Decides:** The image carries no information.

A	Absolute language	Decorative images are the exception.
B	Dropped element	Without alt, some screen readers read the file name.
C	Confusion set	Adds noise for screen-reader users with no benefit.
D	Correct	Decorative images should be ignored by assistive technology.

4. Answer B Concept 3

A designer chose a gray ~~to match the company logo~~ for 16px body text. ~~The page recently won an internal design award.~~ The text-to-background contrast ratio is 3.8:1. Does it meet WCAG Level AA?

Target word: AA pass or fail for body text **Decides:** 16px is normal text, which needs 4.5:1.

A	Adjacent category	7:1 is the stricter AAA level.
B	Correct	3.8:1 is below the AA threshold for normal text.
C	Camouflage	Brand consistency does not change the contrast rule.
D	Confusion set	3:1 applies to large text, not 16px body text.

5. Answer A Concept 3 · transfer

A 24px heading has a contrast ratio of 3.5:1 against its background. Does it meet WCAG Level AA?

Target word: AA for large text — reverse of the body-text item **Decides:** 24px counts as large text, which needs 3:1.

A	Correct	3.5:1 exceeds the 3:1 large-text minimum.
B	Dropped element	Drops the large-text exception.
C	Absolute language	Headings are text and must meet contrast rules.
D	Adjacent category	7:1 is AAA for normal text.

6. Answer A Concept 4

~~A 1,200-line stylesheet contains~~ p { color: black; }, then #intro { color: green; }, ~~and, last in the file,~~ .note { color: blue; }. The element is <p id='intro' class='note'>. What color is its text?

Target word: winning rule **Decides:** The ID selector has the highest specificity.

A	Correct	An ID selector outranks a class and an element selector.
B	Absolute language	The reverse is true.
C	Camouflage	Source order breaks ties only when specificity is equal.
D	Inverted operation	Element selectors have the lowest specificity.

7. Answer C Concept 4 · transfer

An element <p class='note'> is styled by p.note { color: red; } and, later in the file, .note { color: blue; }. What color is its text?

Target word: specificity vs. order **Decides:** p.note (element + class) is more specific than .note.

A	Inverted operation	Author styles override defaults when they match.
B	Confusion set	Treats both selectors as equal in specificity.
C	Correct	Element plus class outranks class alone, regardless of order.
D	Absolute language	Order matters only when specificity ties.

8. Answer B Concept 5

A .card rule sets width: 300px; padding: 20px; border: 2px solid; ~~margin: 16px; background: #ffeccc; and~~ box-sizing: content-box. How wide is the card from its left border edge to its right border edge?

Target word: rendered border-box width **Decides:** Content-box adds padding and border on both sides.

A	Confusion set	Adds padding and border on one side only.
B	Correct	$300 + 20 + 20 + 2 + 2 = 344\text{px}$.
C	Dropped element	Ignores padding and border.
D	Camouflage	Adds the margin, which sits outside the border.

9. Answer D Concept 5 · transfer

A box has width: 300px; padding: 20px; border: 2px solid; and box-sizing: border-box. How wide is its content area?

Target word: solve for content width **Decides:** Border-box fits padding and border inside the 300px.

A	Dropped element	Ignores the padding and border inside the box.
B	Inverted operation	Adds padding and border instead of subtracting (content-box logic).
C	Confusion set	Subtracts padding and border for one side only.
D	Correct	$300 - 40 - 4 = 256\text{px}$.

10. Answer A Concept 6

A site ~~uses a three column grid on desktop and is hosted on a fast content delivery network. It~~ has media queries for small screens, but on phones it still looks like a tiny, zoomed-out desktop page. What is most likely missing?

Target word: missing responsive piece **Decides:** Phones render at a desktop width without the viewport tag.

A	Correct	Without it, mobile browsers render at a desktop width and ignore small-screen queries.
B	Dropped element	The existing queries never trigger without the viewport setting.
C	Inverted operation	Fixed widths make the page less responsive.
D	Camouflage	Speed does not affect layout scaling.

11. Answer B Concept 7

~~A 40-line~~ script in the page's head calls document.getElementById('menu') to attach a click handler. ~~The menu has six links.~~ The call returns null, though the element exists further down the page. What is the best fix?

Target word: fix a null element lookup **Decides:** The script runs before the element is parsed.

A	Absolute language	They can, once the DOM is ready (defer or DOMContentLoaded).
B	Correct	The script then runs after the HTML is parsed.
C	Confusion set	The selector type is not the problem; timing is.
D	Adjacent category	A different lookup still runs before the element exists.

12. Answer C Concept 8

A designer adds outline: none to all buttons and links because the focus ring looks unattractive. ~~User research shows 80% of visitors use a mouse.~~ What should the team do?

Target word: response to removed focus outline **Decides:** Keyboard users need a visible focus indicator.

A	Camouflage	Keyboard users still need to see focus.
B	Inverted operation	Removes keyboard access entirely.
C	Correct	Keeps focus visible while addressing the look.
D	Dropped element	Buttons need visible focus too.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Semantic HTML	Use the element that matches the meaning (button, nav, h1-h6, label); it brings built-in keyboard support and screen-reader roles.	Building controls from div or span and styling them to look right.	A clickable thing made from a generic element.
2	Alt text	Informative images need alt text that conveys their purpose; purely decorative images get an empty alt (alt="").	Leaving alt off, using the file name, or describing decoration.	Whether the image carries information the user needs.
3	WCAG color contrast	Level AA needs at least 4.5:1 for normal text and 3:1 for large text (about 24px, or 18.66px bold).	Using the large-text threshold for body text, or the AAA 7:1 rule for AA.	A contrast ratio and a text size.
4	CSS specificity	Inline > ID > class/attribute/pseudo-class > element; source order breaks ties only between equal specificity.	Thinking the last rule in the file always wins.	One element matched by selectors of different types.
5	The box model	With content-box sizing, rendered width = width + left/right padding + left/right border; with border-box, padding and border fit inside the width.	Forgetting both sides, or adding margin into the box width.	Width, padding, border, and box-sizing given together.
6	Responsive design basics	Responsive layouts need the viewport meta tag plus fluid widths and media queries.	Adding more media queries when phones still show a zoomed-out desktop page.	A site that looks tiny on phones despite media queries.
7	Script timing and the DOM	A script can only find elements that have already been parsed; use defer or place the script at the end of the body.	Blaming the selector when the element simply does not exist yet.	getElementById returns null from a script in the head.
8	Keyboard access and visible focus	Every interactive element must be reachable by keyboard and show a visible focus indicator.	Removing focus outlines because most users have a mouse.	outline: none or tabindex changes on controls.

Course scope (public description): Web Design Fundamentals provides students with essential web development skills, guiding them through the creation of accessible, responsive, and interactive web pages. This course will enable students to gain hands-on experience in HTML, CSS, and foundational JavaScript. The course will also focus on teaching students how to structure content, apply styling, and add dynamic functionality to enhance user engagement. Emphasizing industry best practices, this course covers key aspects of web accessibility, focusing on WCAG compliance, along with core UX and UI principles to ensure usability for diverse audiences. By the end of the course, students will have a comprehensive toolkit for designing web pages that are not only visually compelling but also technically sound and adaptable to various devices and user needs. There are no prerequisites for this course.